Chen 2021 Prostate — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

12/13 samples (≥ 55 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); the whole low-cycling malignant manifold (14,266 cells, cycling 2.1%) carries the trajectory — no separable compartment.

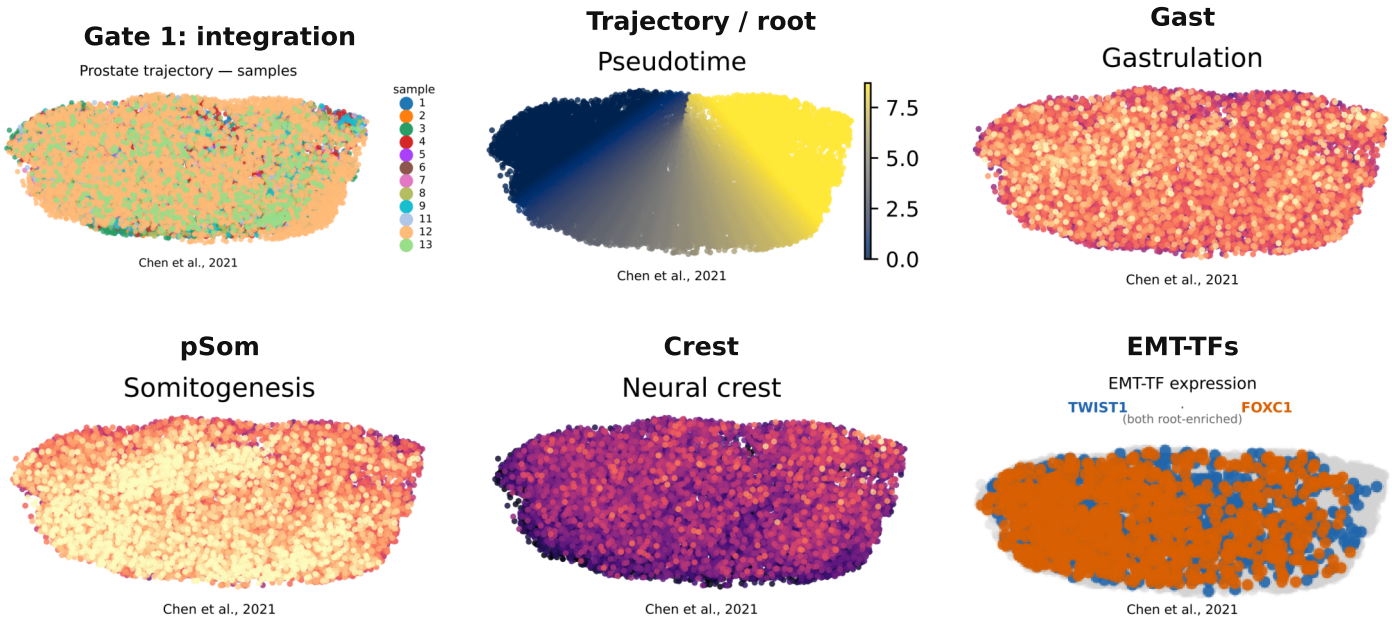
Gate 2 — module geography → GO (non-branching)

Somitogenesis & Gastrulation decline along pseudotime (ρ=−0.32 / −0.18); Neural-crest anti-correlated with pSom (r=−0.76), rising to a terminal readout (ρ=+0.41; crest-high decile mean pt=0.92). Gast/pSom do not co-vary (r=−0.11) — an inverse gradient, no hub.

Gate 3 — hub mutual exclusivity → NA

Single monotone pSom→Crest axis — no bifurcation and no Gast/pSom co-expressing hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES



Dataset. Prostate cancer (Chen et al., 2021), 10x — 13 samples / 36,424 cells (master scope); malignant only: 14,291 load, 14,266 across 12 samples (≥ 55) enter preprocessing; all 14,266 (one continuous manifold, 0 detached islands) enter the trajectory.

Verdict. ✓ Non-branching pattern — Gate 1 GO, Gate 2 GO, Gate 3 NA. Chen's prostate malignant manifold is a single monotone Somitogenesis → terminal Neural-crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Expressed EMT-TFs (TWIST1, FOXC1) are root-enriched and decline toward the terminus — no canonical EMT-TF is induced at the neural-crest end; SNAI1/ZEB1 near the detection floor (3.0% / 0.4%).

Parameter choices

Min cells/sample	≥ 55 (keeps 12/13; gap 125→25)	**Integration**	Harmony θ=10; PCA 50PC; spread=1.3
Compartment	whole manifold (2.1% cycling, no split)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Orientation**	x-flip, no swap (raw-x ‖ pt, r=+0.94)